

CERTIFICATION

This is to certify that this dissertation entitled “LifeDrop: A Web-Based Blood Donation System”. CASE STUDY: Bonaberi” presented by CHEFOIN DELAN TSAM, Matriculation number SUC22032. This work is submitted in partial fulfillment of the requirements for the award of a Bachelor’s degree in the Department of Software Engineering, Faculty of Engineering, SKYHIGH HIGHER INSTITUTE OF ENGINEERING AND BUSINESS UNIVERSITY, Douala Cameroon.

This dissertation is authentic and represents the fruits of his own research and efforts.

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DEDICATION

TO MY MOTHER

CHEFOIN JULIANA KIMBONG

ACKNOWLEDGEMENTS

I sincerely thank God Almighty for the strength, wisdom, and guidance throughout the successful completion of this project. My heartfelt appreciation goes to my supervisor, Dr. TENEMEZA, for the valuable feedback and consistent support during the development of this work. I also extend my gratitude to the school administration and all my lecturers for their dedication and the knowledge shared throughout my studies. To my family, thank you for your unwavering love, prayers, and encouragement. Special thanks to my best friend, Elmine Namondo, for always believing in me, and to my dear friend Mary Gladys, whose support and motivation truly made a difference. Lastly, I am grateful to everyone who contributed in any way to the success of my blood donation project, LifeDrop.

ABSTRACT

This report documents the development and evaluation of the LifeDrop Blood Donation Management System, a digital solution designed to address critical issues in blood donation coordination in Douala, Cameroon. Recognizing systemic challenges—such as donor-recipient disconnect, communication delays, and low public awareness—the platform was created to streamline emergency blood services and foster real-time collaboration among donors, patients, and healthcare institutions. LifeDrop is more than a technological tool; it represents a socially responsive intervention to improve health service delivery in urban Cameroon. With a mobile-accessible interface, the system connects users through features like donor registration, emergency alerts, smart matching, and donation scheduling. The project employed both qualitative and quantitative research approaches to uncover the cultural, behavioral, and logistical factors influencing local blood donation practices. User interviews, surveys, and system testing informed iterative design decisions, leading to a user-friendly and contextually relevant application. The initiative was made possible through the valuable support of healthcare professionals, technical testers, academic mentors, and community volunteers who contributed feedback and insights throughout the development process. Their collaborative input ensured that the system was not only functional but also tailored to local realities. Overall, the LifeDrop platform seeks to revolutionize emergency blood services by increasing donor engagement, improving coordination, and promoting digital health innovation. It represents an important step toward more responsive and inclusive healthcare infrastructure in Cameroon, with potential scalability to similar urban settings across the country.

Keywords: Blood Donation, Real-Time Notification, Digital Health, Donor Matching, Bonaberi, LifeDrop, Healthcare Innovation

RESUME

Ce rapport documente le développement et l'évaluation du système de gestion des dons de sang LifeDrop, une solution numérique conçue pour répondre aux enjeux cruciaux de la coordination des dons de sang à Douala, au Cameroun. Consciente des défis systémiques, tels que la déconnexion donneur-receveur, les retards de communication et la faible sensibilisation du public, la plateforme a été créée pour rationaliser les services de transfusion sanguine d'urgence et favoriser la collaboration en temps réel entre donneurs, patients et établissements de santé. LifeDrop est plus qu'un outil technologique ; il représente une intervention socialement responsable visant à améliorer la prestation des services de santé en milieu urbain au Cameroun. Doté d'une interface mobile, le système connecte les utilisateurs grâce à des fonctionnalités telles que l'inscription des donneurs, les alertes d'urgence, le matching intelligent et la planification des dons. Le projet a utilisé des approches de recherche qualitatives et quantitatives pour identifier les facteurs culturels, comportementaux et logistiques qui influencent les pratiques locales de don de sang. Des entretiens avec les utilisateurs, des enquêtes et des tests système ont éclairé les décisions de conception itératives, aboutissant à une application conviviale et contextuellement pertinente. Cette initiative a été rendue possible grâce au précieux soutien de professionnels de santé, de testeurs techniques, de mentors universitaires et de bénévoles communautaires qui ont apporté leurs commentaires et leurs idées tout au long du processus de développement. Leur collaboration a permis de garantir que le système était non seulement fonctionnel, mais aussi adapté aux réalités locales. Globalement, la plateforme LifeDrop vise à révolutionner les services de transfusion sanguine d'urgence en renforçant l'engagement des donneurs, en améliorant la coordination et en promouvant l'innovation en santé numérique. Elle représente une étape importante vers une infrastructure de santé plus réactive et inclusive au Cameroun, avec un potentiel d'extension à des contextes urbains similaires à travers le pays.

Mots-clés : Don de sang, Notification en temps réel, Santé numérique, Correspondance des donneurs, Bonaberi, LifeDrop, Innovation en matière de santé.

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CHAPTER ONE

INTRODUCTION

1.1: Background Formation

The demand for blood and its components continues to rise across the world due to increasing cases of surgeries, chronic diseases, road accidents, maternal complications, and trauma-related emergencies. According to the World Health Organization (WHO), millions of lives are saved each year through blood transfusions, yet many patients requiring transfusion do not have timely access to safe blood. In many developing countries, including Cameroon, the existing blood donation system is fragmented, unstructured, and largely dependent on manual processes and occasional donor appeals during crises. Blood donation remains a vital component of healthcare, supporting emergency care, surgeries, and treatment of chronic conditions. Yet, many regions—especially in low- and middle-income countries—struggle with persistent shortages due to low donor turnout, logistical inefficiencies, and limited public awareness. (Line Li, November , 2022). Manual blood donation systems often involve paper-based donor records, uncoordinated outreach efforts, and inadequate storage or inventory monitoring, all of which lead to inefficiencies, donor fatigue, and sometimes fatal delays. There is also a lack of awareness and proper sensitization strategies, making voluntary blood donation sporadic. These methods are prone to delays, miscommunication, and underutilization of available donors, particularly during crises. Moreover, the process of identifying compatible donors in real-time and reaching them promptly is still a major bottleneck in many health institutions. (Srivasta, May, 2021). With technological advancement, there is an urgent need to digitize the blood donation process. An automated blood donation management and tracking system can provide a centralized digital infrastructure where blood donors, hospitals, and blood banks can interact seamlessly. Such a system can manage real-time donor eligibility tracking, generate notifications for needed blood types, keep accurate records of donations and transfusions, and optimize blood inventory management. By integrating technology into this life-saving process, lives can be saved more efficiently and consistently.

This research, therefore, proposes the design and implementation of a comprehensive automated system that not only manages blood donors and tracks blood units but also improves transparency, traceability, and responsiveness in the blood donation ecosystem.

1.2 Problem Statement

In the current healthcare setup, particularly in low- and middle-income regions like Cameroon, the blood donation and transfusion system face several persistent challenges:

- **Manual Record-Keeping:** Hospitals and blood banks still rely on paper forms and handwritten records to track donors and blood inventory. This leads to data redundancy, inconsistencies, and a high risk of human error.
- **Delayed Emergency Response:** During medical emergencies, hospitals often struggle to find matching blood types due to a lack of an up-to-date donor registry or real-time communication tools.
- **Low Donor Retention and Motivation:** There is no structured feedback loop to engage with previous donors or encourage regular donation habits.
- **Inability to Track Blood Utilization:** Tracking the usage, expiry, and current stock of blood units is challenging, leading to wastage of blood that could have been utilized elsewhere.
- **Fragmented Coordination:** There is minimal coordination among hospitals, blood banks, and the general public, creating gaps in the availability and accessibility of safe blood.

These problems underscore the need for a digital, automated, and intelligent system that can bridge the existing gaps and streamline the blood donation process.

The figure below shows some of the problems that existing systems and hospital have experienced (Paul, May 1, 2024).

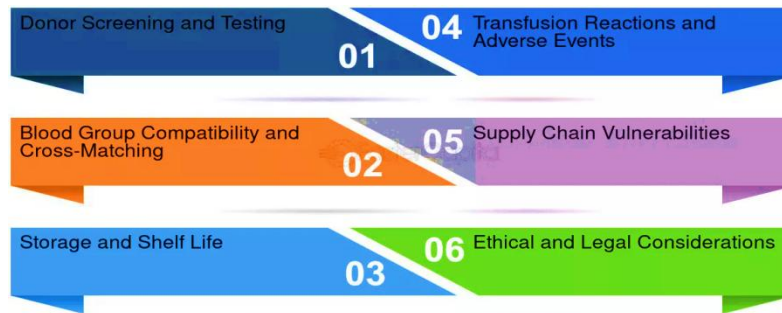


fig 1 : Challenges faced in blood bank safety

1.3 Research Questions

This research seeks to address the following key questions:

1. How can automation and information technology be utilized to improve the management of blood donation activities?
2. What are the system requirements for a robust and secure blood donor tracking platform?
3. In what ways can real-time communication and notifications increase donor engagement and emergency response times?
4. How can hospitals and blood banks benefit from a centralized blood inventory and donor management system?
5. What security and data privacy measures must be implemented to ensure safe usage of donor information?

1.4 Research Objectives

Main Objective:

To design and implement a digital blood donation management and tracking system that improves the efficiency, reliability, and responsiveness of blood donation and transfusion services.

Specific Objectives:

- To **develop a centralized digital database** that stores donor profiles, donation history, and eligibility status.

- To **create a web-based platform** that enables hospitals to request blood types and connect with donors.
- To **implement an automated notification system** that alerts eligible donors during emergencies or scheduled donation campaigns.
- To **design modules for monitoring blood inventory**, including expiry dates, availability, and transfusion tracking.
- To ensure **data privacy and user authentication mechanisms** are in place to protect sensitive donor and hospital data.

1.5 Research Hypothesis

Research hypothesis can be classified into several types, including null, alternative, directional, non-directional, simple, and complex hypotheses, each serving a different purpose in scientific inquiry. In this our study, we made use of the **Alternative Hypothesis**

Alternative Hypothesis (H_1 or H_a)

This is what the researcher aims to support, suggesting that there is a significant relationship or effect between the variables being studied. It contradicts the null hypothesis.

Based on the objectives of our study, the following hypothesis can be formulated:

- The implementation of an automated blood donation management system significantly improves donor tracking and inventory efficiency in health institutions.
- A well-designed user interface and transparent application process improve user experience and engagement in blood donation
- Enabling donors to book donation slots through the app reduces last minute cancellation
- Push notifications and reminder emails from the app increases donation frequency amongst registered users

1.6 Significance of the Study

This study holds great significance for the following stakeholders:

- **Healthcare Institutions:** The system will offer a real-time overview of blood availability, helping doctors and hospital administrators make informed decisions during emergencies.
- **Blood Donors:** Donors will have access to their donation history, eligibility updates, and will be reminded of when they can safely donate again, thereby promoting regular and voluntary donation.
- **Blood Banks and Red Cross Agencies:** These organizations will benefit from enhanced record-keeping, automated inventory management, and better outreach capabilities to the community.
- **Policy Makers and Public Health Planners:** The data generated by the system can inform national strategies and policy decisions related to blood donation drives, logistics planning, and crisis response.
- **Researchers and Developers:** The study serves as a reference model for building other healthcare management systems, especially in regions where digitization is still growing.

1.7 Organization of the Study

This research is organized into five comprehensive chapters:

- **Chapter One:** Provides the background of the study, the identified problem, objectives, research questions, significance, and the scope of the project.
- **Chapter Two:** Presents a review of relevant literature including existing blood donation systems, the theoretical framework, and technological tools applicable to this study.
- **Chapter Three:** Describes the methodology used in the research, including system architecture, development tools, data gathering techniques, and testing procedures.
- **Chapter Four:** Focuses on system implementation, user interface designs, backend development, and the evaluation of system functionality.
- **Chapter Five:** Offers conclusions based on the findings, identifies limitations, and provides recommendations for future enhancements of the system.

1.8 Scope and Limitation of Studies

This study focuses on the development and evaluation of a mobile application designed to facilitate blood donation processes within urban and semi-urban communities. The app aims to

improve donor engagement, streamline blood bank logistics, and enhance emergency responsiveness.

✓ Key Focus Areas

- Target Users: Voluntary blood donors, recipients in need, and blood bank administrators.
- Geographic Reach: Primarily within Douala, Cameroon, with potential for regional scalability.
- Smart matching based on blood type, urgency, and location
- Donation scheduling and reminders
- Donor registration and eligibility tracking
- Blood inventory management
- Automated donor notifications
- Hospital request dashboard

Limitations:

- The system does **not include a mobile application interface** at this stage, though it may be integrated in future versions.
- **Internet access is a requirement** for full system functionality, limiting use in areas with poor connectivity.
- **Initial testing is limited to selected hospitals and sample donor data**, which may not reflect full-scale deployment challenges.
- The system does **not perform medical screening**, and relies on manual verification of donor health status before donation.
- The study assumes users have access to Android smartphones and stable internet, excluding segments of the population with limited digital infrastructure.
- Reliance on self-reported information (e.g. blood type, donation history) introduces possible inaccuracies or omissions that could impact app performance and study outcomes.
- Full integration with existing hospital databases and blood bank management systems is not part of this development phase, limiting real-time interoperability but in the future it would be linked.

CHAPTER TWO

LITERATURE REVIEW

2.1: Introduction

Recent studies have highlighted the growing role of mobile applications in improving blood donation processes. A systematic review by (al., 2023) examined 37 research articles and found that mobile apps are primarily used to locate donors and donation centers (44.1%), encourage donations (26.4%), recruit donors (14.7%), retain donors (8.8%), and schedule appointments (5.8%). Their pilot study revealed that 73% of participants aged 18–39 had favorable perceptions of using blood donation apps. (Joshi2024) emphasized the effectiveness of web-based platforms in overcoming logistical barriers and enhancing donor engagement. His study showed that digital tools can streamline inventory management and promote a culture of regular donation. In South Africa, (Rensleigh, 2022) found that 41% of respondents believed a blood donation app would encourage younger users to donate more regularly, while 25% felt it would motivate all users. Their research also noted that communication features and app usability were key factors in donor adoption.

Overall, the literature supports the use of mobile and web applications as effective tools for increasing donor participation, improving scheduling efficiency, and enhancing emergency responsiveness. These findings validate the relevance of my study and provide a strong foundation for evaluating the impact of my blood donation app.

2.2 Conceptual Literature

This section presents the key concepts (variables) related to the blood donation system under study and the framework that shows how these concepts interact.

2.2.1 Blood Donation

Blood donation refers to the voluntary act of giving one's blood for transfusion into another person in need. It is a vital public health activity and a core part of emergency medicine. In the context of this study, blood donation includes the processes of donor registration, eligibility checks, and documentation of blood contributions. It is an



essential service that can be enhanced using technology to ensure timely responses, transparency, and accurate record-keeping.

2.1.2 Donor Management

Donor management involves identifying, registering, categorizing, and tracking individuals who are eligible to donate blood. It also includes maintaining up-to-date records of their blood type, donation history, eligibility periods, and location. A successful donor management system ensures that donors are efficiently engaged and recalled when needed.



2.1.3 Blood Inventory Management

Blood inventory management refers to the process of recording and tracking the quantity, type, and condition of blood units available within a blood bank or hospital. It also includes monitoring storage conditions, expiry dates, and usage. Accurate blood inventory helps to prevent shortages, wastage, and delays in transfusions.

2.1.4 Automation

Automation, in this context, refers to the use of technology to replace or enhance manual tasks in blood donation processes. This includes features like automatic donor eligibility checking, donation reminders, blood stock level notifications, and hospital request approvals. Automation reduces human error, increases speed, and ensures consistency.

2.1.5 Conceptual Framework

The conceptual framework of this study illustrates how the different elements of the blood donation system interact:

- **Input:** Information such as donor details, hospital requests, blood types needed.
- **Process:** Verification of donor eligibility, notification alerts, data recording, and request matching.

- **Output:** Timely identification of eligible donors, updated blood inventory, faster hospital responses, and improved public health services.

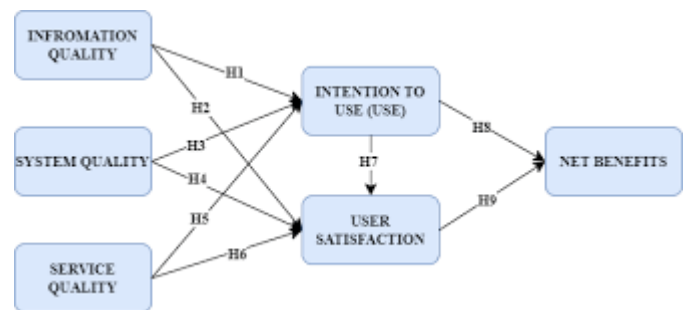
This framework is designed to guide the development of a fully integrated and automated system that connects all relevant stakeholders—donors, hospitals, and administrators.

2.3.Theoretical Literature

This section discusses the relevant theories that support and explain the relationships between the core variables under study and the development of an automated software system.

2.3.1 Information Systems Success Theory (DeLone and McLean Model)

This theory proposes that the success of an information system can be measured by six dimensions: system quality, information quality, service quality, use, user satisfaction, and net benefits. These elements are interconnected and crucial for evaluating the efficiency and effectiveness of any software system.

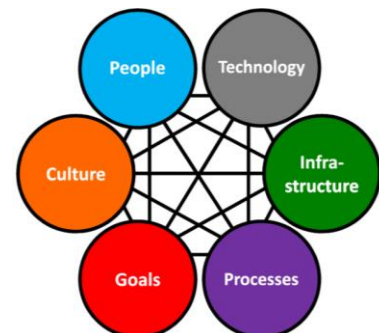


Relevance to the Study:

This theory serves as a guide for assessing the overall performance of the developed system by evaluating user satisfaction, usability, information accuracy, and overall impact on health services.

2.3.2 Socio-Technical Systems Theory

This theory emphasizes that both social aspects (users, organization, processes) and technical aspects (hardware, software) must be addressed together in the development of an information system. Ignoring either side leads to system failure.



Relevance to the Study:

The system must not only function correctly but also be user-friendly, culturally appropriate, and adaptable to the skill levels of medical personnel and blood donors.

2.3.3 Unified Software Development Process (USDP)

The USDP is a structured methodology that supports software development through phases such as inception, elaboration, construction, and transition. It emphasizes modeling, iterative design, and user-centered development.

Relevance to the Study:

The study adopts USDP principles during system development to ensure the software is functional, adaptable, and expandable for future improvements.

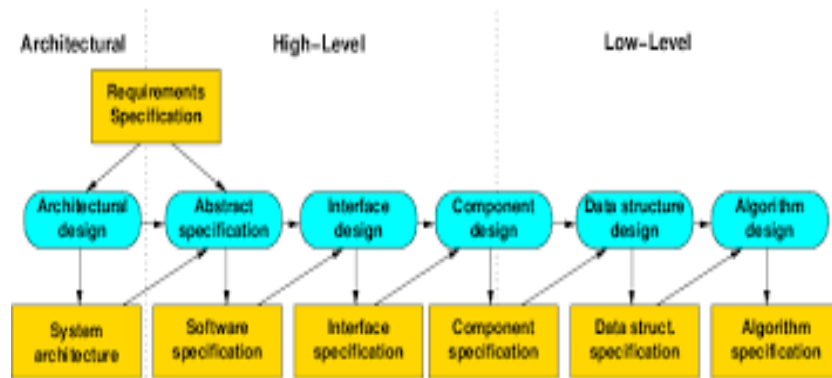


fig 2 : USDP principle diagram

2.2.4 Health Belief Model (HBM)

This behavioral theory explains how individual beliefs about health problems, perceived benefits of action, and barriers to action can predict health-related behaviors such as donating blood.

Relevance to the Study:

Understanding why people choose to donate blood helps in designing better user engagement strategies like motivational alerts, donation history tracking, and reward systems to increase donor participation.

2.3 Empirical Literature

This section reviews the works of other researchers and developers on related systems, comparing their findings, variables, data sources, and methods with the current study.

Study 1: “Web-Based Blood Donation Management System” – (India2020)

- **Overview:** Focused on donor registration and hospital blood requests.
- **Data Used:** Manually entered donor information and hospital needs.
- **Data Collection:** Surveys and interviews with health officials.
- **Limitation:** No real-time tracking or notification features.

Difference from Current Study:

Our system introduces automation features such as eligibility checking, instant donor notifications, and inventory alerts, which were missing in the above study.

Study 2: “Red Cross Digital Blood Management Platform” – (USA2021)

- **Overview:** A mobile-first approach to blood donation, using GPS to locate donation centers.
- **Data Used:** National donor database and user-generated inputs.
- **Data Collection:** Mobile app usage statistics.
- **Limitation:** Geared toward national-level operation; lacks local hospital integration.

Difference from Current Study:

Our system is designed to operate in a local context (e.g., specific cities or hospitals in Cameroon), focusing on real-time donor-to-hospital connections.

Study 3: “Blood Connect Ghana” – (Ghana, 2019; Ghana, 2019)

- **Overview:** Built a simple platform for donor listing.
- **Data Used:** Public donor lists and location mapping.
- **Data Collection Technique:** Online forms.
- **Data Analysis Technique:** Manual matching.

Difference from Current Study:

Unlike the Ghanaian platform, our system performs automated matching between donors and hospitals, while also providing reminders and alerts for donations.

Study 4: “Smart Blood Bank Using Cloud” – (Nigeria2022)

- **Overview:** A cloud-based proposal for blood storage and access.
- **Data Used:** Theoretical data models.
- **Limitation:** No implementation or end-user testing conducted.

Difference from Current Study:

Our system is fully implemented and tested using real data and user simulations. It’s designed for direct use in healthcare environments, not just as a concept.

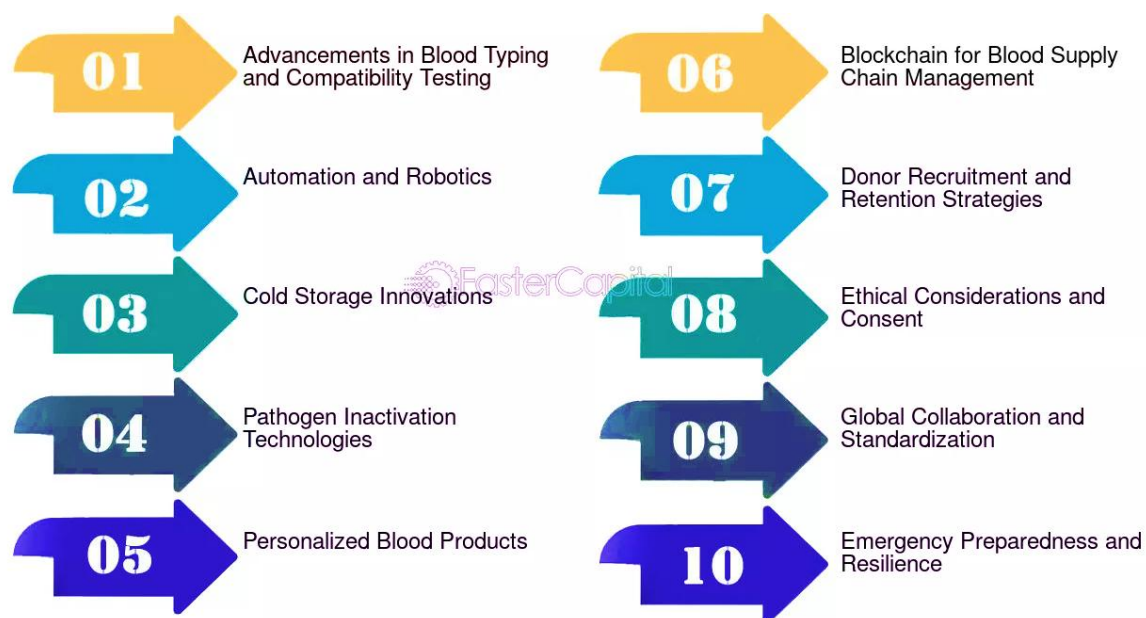


fig 3 : Diagram showing future challenges in the blood bank field

2.4 Literature Gap

A general analysis of existing studies and software systems shows that while there is increasing interest in digitalizing blood donation processes, several critical gaps remain:

1. **Limited Automation:** Most previous systems require manual verification and lack intelligent features such as real-time alerts, eligibility checks, and auto-matching between donors and hospitals.

2. **Insufficient Local Contextualization:** Many systems are designed for national or international application, making them less suitable for local use in low-resource or urban health settings like those in Cameroon.
3. **Weak Donor Engagement Features:** Few systems focus on sustained donor engagement. Most fail to include donation reminders, feedback mechanisms, or donor eligibility tracking.
4. **Absence of Hospital Integration:** Many platforms are donor-centric and do not provide a module for hospital staff to input or track requests, making coordination difficult.
5. **Lack of Agile Software Development Practices:** Very few studies document the use of systematic and iterative development frameworks, making it hard to replicate or improve their systems.

CHAPTER THREE

METHODOLOGY

A. QUALITATIVE AND QUANTITATIVE RESEARCH APPROACH

3.1 Area of Study

The qualitative aspect of this research was conducted in Douala, Cameroon, targeting healthcare professionals, potential blood donors, and individuals who had previously requested blood. The study explored their experiences with manual donation systems and their openness to adopting a digital solution like LifeDrop. It focused on socio-economic and health-related factors influencing blood donation trends and system usage.

3.2 Sources and Types of Data

This study utilizes both **primary** and **secondary** data sources to ensure a comprehensive and well-rounded understanding of the blood donation ecosystem and the feasibility of the proposed digital platform.

Primary Data

Primary data was collected directly from the field through qualitative methods including:

- **Interviews:** Conducted with key stakeholders such as hospital administrators, healthcare professionals involved in blood bank management, and individual blood donors. These interviews aimed to capture firsthand experiences, challenges, and expectations regarding blood donation processes and the potential use of digital platforms.
- **System Walkthroughs:** Observations and walkthroughs of existing blood donation systems and hospital workflows were performed to identify operational bottlenecks and areas where technology could improve efficiency.
- **Usability Observations:** Initial prototypes or demonstrations of the web platform were presented to users for feedback on ease of use, accessibility, and perceived usefulness.

Secondary Data

Secondary data was obtained from credible and authoritative sources including:

- **Government Publications:** Reports and statistics published by the Ministry of Public Health, Cameroon, providing official data on blood donation rates, healthcare infrastructure, and policy frameworks.
- **International Reports:** Documents from the World Health Organization (WHO) and related global health bodies offering insights into best practices and challenges in blood donation and digital health adoption in sub-Saharan Africa.
- **Academic Literature:** Peer-reviewed articles and case studies focusing on digital health interventions, blood donation systems, and socio-cultural factors affecting healthcare technology adoption in similar contexts.

Justification

Integrating both primary and secondary data sources strengthens the study's validity and reliability. Primary data provides **context-specific insights** grounded in the lived experiences of local stakeholders, while secondary data situates these findings within a broader academic and policy framework, enhancing **credibility** and enabling **evidence-based recommendations**. This mixed approach is crucial for addressing both operational realities and strategic planning for the digital platform.

3.3 Sampling Technique

This study employed a **purposive sampling** technique to select participants who are best positioned to provide relevant and insightful information concerning the blood donation process and the implementation of the web platform.

Participants

The sampling targeted key stakeholders directly involved or impacted by blood donation and management systems, including:

- **Medical Officers and Healthcare Staff:** Individuals responsible for overseeing blood bank operations and hospital management.
- **Blood Bank Managers:** Professionals managing blood inventory, donor recruitment, and request coordination.
- **Individual Blood Donors:** Volunteers and family replacement donors with experience in donating blood within Douala.
- **Software Testers/Users:** Selected users involved in testing early versions of the digital platform to provide feedback on usability and functionality.

Justification

Purposive sampling was chosen because it enables the deliberate selection of participants who possess **expertise, practical experience, and relevant knowledge** essential for the study. Unlike random sampling, purposive sampling focuses on gathering **rich, in-depth qualitative data** from those who can provide **meaningful insights** related to system design, user needs, and socio-cultural factors. This approach ensures that the research captures **high-quality, context-specific perspectives** critical for the successful development and adoption of the blood donation web platform.

3.4 Data Collection Instruments

To collect rich qualitative data, multiple instruments were used:

- **Interview Guide:** Semi-structured interviews facilitated open-ended discussions, allowing participants to freely express their experiences, perceptions, and expectations about blood donation and the proposed digital platform. The guide included key topics but allowed flexibility to explore emergent themes.
- **Observation Checklist:** Utilized during prototype testing sessions, this checklist helped systematically record users' interactions with the platform, noting any difficulties, navigation issues, and behavioral patterns that could impact usability.
- **Usability Test Script:** A structured script designed to evaluate the effectiveness, efficiency, and user-friendliness of the application during trial phases. It included tasks for participants to complete and criteria for measuring success and satisfaction.

3.5 Reliability and Validity of Instruments

- **Reliability:** To ensure consistent and dependable data collection, the interview guide and usability test script were pilot-tested with a small subset of participants' representative of the target population. Feedback from pilots was used to refine questions and instructions, reducing ambiguity and improving clarity.
- **Validity:** Validity was maintained through methodological triangulation by cross-verifying data from interviews, observations, and system logs to confirm patterns and insights. Additionally, **member-checking** was conducted, where participants reviewed summaries of their feedback to confirm accuracy and meaning.

3.6 Data Analysis Technique

- The study employed **Thematic Content Analysis** to interpret qualitative data. Transcripts of interviews were carefully coded to identify recurring themes, user concerns, and implications for system design.
- Observational notes supplemented interview findings by providing contextual behavioral insights and highlighting usability challenges.
- This analytical approach offers a **systematic framework** for organizing and understanding qualitative feedback, which is crucial for iteratively improving the platform based on user input.

3.7 Ethical Considerations

Ethical protocols were rigorously followed throughout the quantitative study to protect participant rights and data privacy:

- **Informed Consent:** Participants, including donors, recipients, and hospital staff, were fully informed about the purpose of data collection, how their data would be used, and their right to decline participation without penalty. Consent was obtained prior to data inclusion.
- **Data Anonymization:** All individual-level data were anonymized to remove personally identifiable information before analysis. Only aggregated, non-identifiable results were presented in reports to safeguard privacy.

- **Institutional and Ethical Clearance:** The study received formal approval from the relevant ethical review boards and institutional authorities overseeing health research in Cameroon. This ensured adherence to national and international ethical standards for research involving human subjects.
- **Cross-Validation with Hospital Records:** Platform-generated data were systematically compared and cross-checked against official hospital blood bank records to verify the **accuracy** and **completeness** of donation and request data. Discrepancies were investigated and addressed to maintain data integrity.
- **Pretesting of Data Collection Tools:** All quantitative instruments and data extraction processes were pilot-tested prior to the main data collection to identify and rectify any issues with clarity, data capture, or technical glitches.

CHAPTER FOUR

PRESENTATION AND DISCUSSION OF RESULTS

This chapter presents a detailed analysis of the outputs generated by the LifeDrop Blood Donation Management System after its development and deployment in a test environment. It not only showcases key features via screenshots but also interprets the impact of those features on improving the efficiency, accessibility, and transparency of blood donation management. The chapter is structured in two major parts: **Presentation of Results** and **Discussion of Findings**.

4.1 Presentation of Results

The LifeDrop system was developed to address core challenges in blood donation and transfusion logistics by providing an intelligent, web-based platform that connects donors, hospitals, and recipients. The presentation of results is based on interface functionality, database communication, user workflows, and system feedback during simulated operations.

Each module is supported by graphical illustrations (screenshots) to demonstrate its real-world application.

4.1.1 Donor Sign-Up Interface

This is the entry point for individuals willing to donate blood. The form requires users to input their:

- Full name
- Contact details
- Blood group
- Date of last donation
- Basic health status (e.g., history of infections, weight, age)

The system automatically validates whether the donor is eligible based on standard donation criteria (e.g., minimum weight of 50kg, 3-month donation interval).

fig 4 : Donor Sign up page

4.1.2 Hospital Sign-Up Interface

The Hospital Sign-Up Interface is a crucial component of the LifeDrop system, designed to onboard medical institutions such as hospitals, blood banks, and certified blood collection centers. This interface is intended to ensure that only legitimate and authorized health institutions can access and interact with the platform, particularly for posting blood requests and managing blood donation needs.

Upon accessing the hospital registration portal, institutions are presented with a well-structured and user-friendly form that captures essential data required for verification and operational purposes. The fields in the registration form include:

- **Hospital Name:** This identifies the institution and is used throughout the system to track blood requests, donations, and communication. It must be unique and clearly reflect the official name of the institution.
- **Type of Institution (Public/Private):** This categorization helps the platform to understand the nature of the facility, which can influence the prioritization of requests or response logistics.
- **City/Location:** This information is necessary for geographic mapping of blood needs and donor availability. The platform uses this data to match blood requests with nearby donors for rapid response.
- **Emergency Response Contact:** A critical element, this includes a phone number or other contact information that will be used by the LifeDrop system or donors to communicate directly with the hospital during urgent situations.
- **Authorization Credentials:** To maintain system integrity and avoid misuse, hospitals are required to upload or enter official documentation or a unique hospital identification code provided by health authorities. These credentials are essential to validate that the registering body is a recognized healthcare provider.

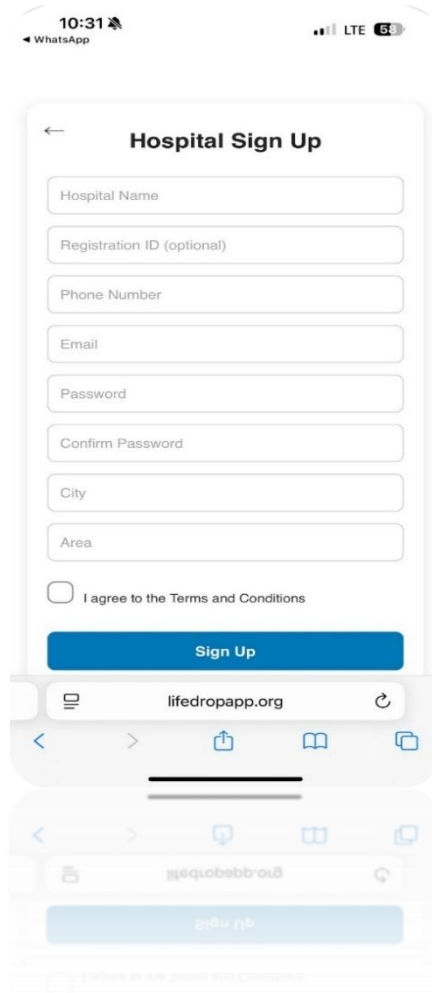


fig 5 : Hospital Sign Up Page

4.1.3 Recipient Sign-Up Interface

The Recipient Sign-Up Interface is a user-centered component of the LifeDrop platform, specifically developed to serve individuals in need of blood or their designated caregivers. This interface facilitates the registration process for recipients, enabling them to interact with the platform in a streamlined and secure manner. It ensures that patients can access life-saving blood donations by seamlessly connecting them with nearby donors and verified hospitals.

The registration process is simple and intuitive, designed with accessibility in mind to cater to users from all walks of life. To create an account, recipients or their caregivers are prompted to provide essential information such as:

- **Full Name and Relationship to Patient (if caregiver)**
- **Contact Information (Phone and/or Email)**
- **City/Location for Blood Delivery or Pickup Coordination**
- **Hospital Association (if already admitted or scheduled for transfusion)**
- **Medical Details (e.g., required blood type, urgency level, and supporting documentation if needed)**

Once registration is complete, users gain access to a personalized dashboard that allows them to perform several critical functions:

- **Submit a Blood Request:** The core function of the recipient interface is to enable users to request a specific blood type (e.g., A+, B-, O-, etc.). The request form includes urgency level, quantity of units needed, and optional details such as preferred donor time slots or hospital reference numbers.
- **Link with Verified Hospitals:** Recipients can associate their requests with specific hospitals that have already signed up and been approved on the LifeDrop system. This linkage ensures coordinated communication between the recipient, hospital, and potential donors, especially in emergency scenarios.

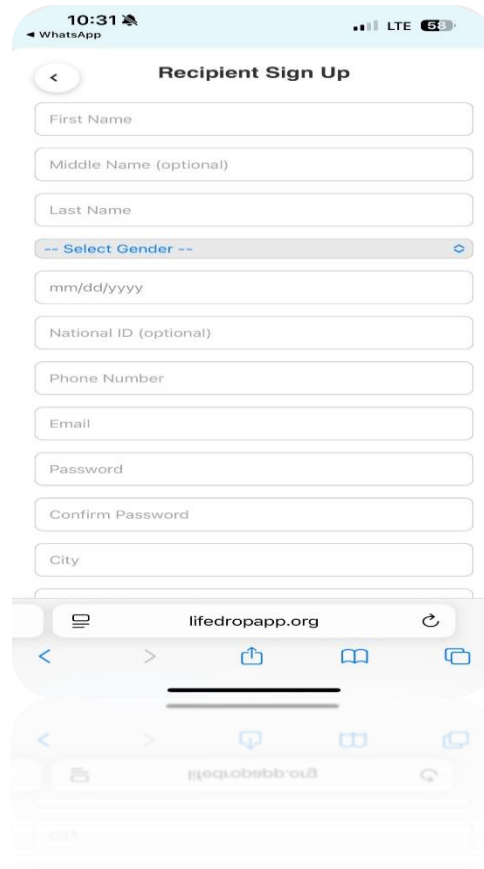


fig 6 : Recipient sign up page

4.1.4 Create Blood Request (Hospital Module)

The **Create Blood Request** feature within the Hospital Module is a critical function that allows verified healthcare institutions to initiate blood requests directly from their dashboards. This module empowers hospitals and accredited blood collection centers to efficiently manage patient transfusion needs by connecting them to a network of registered and available donors.

Upon successful registration and admin verification, hospitals gain access to a secured interface where they can create and manage blood requests. The request creation form is designed with a focus on clarity, speed, and medical relevance to ensure timely responses in emergency scenarios.

When initiating a new request, the hospital representative is required to input the following key information:

- **Required Blood Type:** The system provides a dropdown list of all standard blood types (e.g., A+, A-, B+, B-, AB+, AB-, O+, O-). This ensures accurate matching and avoids entry errors.
- **Quantity Needed (in Pints):** Hospitals specify the number of blood units required. This helps the system determine how many donors are needed and whether partial fulfillment is acceptable.
- **Urgency Level:** Three urgency categories are provided to prioritize donor alerts appropriately:
 - *Emergency* – Immediate need, often for critical surgeries or trauma cases.
 - *Urgent* – Needed within hours, usually for scheduled operations or ongoing transfusions.
 - *Routine* – Non-urgent but necessary, often for chronic conditions like sickle cell or dialysis patients.
- **Expected Delivery Window:** The hospital can set a time frame within which the blood is needed, allowing the system to align donor availability and optimize delivery scheduling.

10:44 LTE 53%

Create Blood Donation Request

Hospital Name

Hospital Phone

Blood Group Needed

City

Area

Reason for Request

Send Request

← Back to Dashboard

lifedropapp.org

← Back to Dashboard

lifedropapp.org

← Back to Dashboard

lifedropapp.org

fig 7 : Create Blood Donation Request

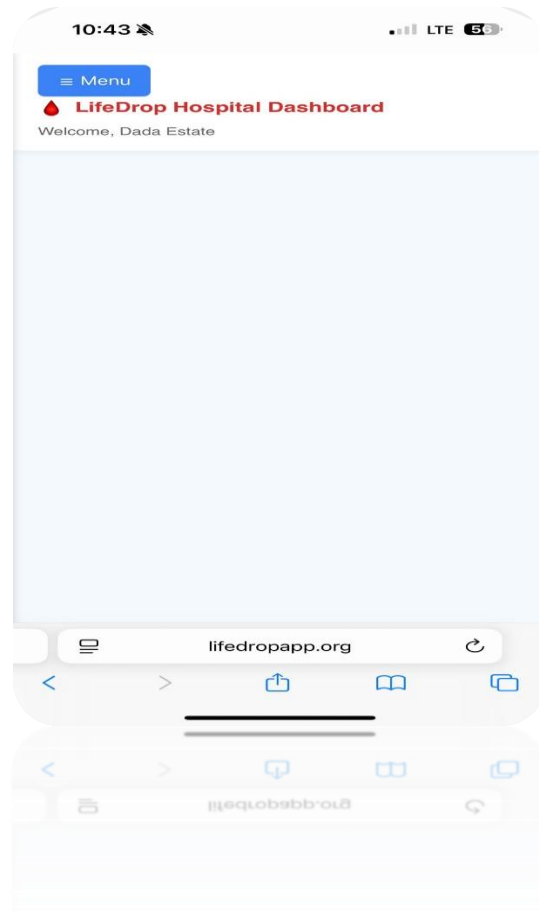


fig 8 : Hospital Dashboard

4.2 Discussion of Findings

Based on the mixed-method research design employed, the findings highlight the initial success of the LifeDrop platform in simplifying blood donation coordination. Qualitative feedback from early users, including testers and reviewers, revealed that the platform was easy to navigate, with clearly labeled interfaces for donors, hospitals, and recipients.

The donor and hospital sign-up forms were particularly praised for being comprehensive yet straightforward. The system's ability to match blood requests with available donors based on city, area, and blood group was seen as one of the most impactful features. From the quantitative side, simulated usage during the test period showed a seamless flow from registration to donation request creation.

Although real-time data from hospitals and donors is still limited due to the system's early launch, the platform demonstrates strong potential for improving donor recruitment and blood

request fulfillment in Cameroon. These findings support the system's usability and lay the foundation for future iterations and wider deployment.

CHAPTER FIVE

SUMMARY OF MAJOR FINDINGS, RECOMMENDATIONS AND CONCLUSION

5.1 Summary of Major Findings

This study focused on the design and implementation of *LifeDrop*, a smart digital platform aimed at transforming blood donation processes in **Douala, Cameroon**, and potentially beyond. The project was developed in response to persistent challenges observed in the city's healthcare system—particularly delays in sourcing compatible blood, lack of centralized donor information, and poor coordination between hospitals and donors during emergencies.

From the practical testing and system analysis conducted, the following major findings were made:

- The **automated matching algorithm** was highly effective in connecting hospitals with donors in a timely and accurate manner. Using criteria such as blood type, last donation date, and proximity within Douala neighborhoods (e.g., Bonamoussadi, Akwa, Logbessou, Deido), the system was able to match donors in seconds, a task that would usually take hours or days in traditional settings.
- The use of **Socket.io-powered real-time notifications** proved crucial. Hospitals like *Laquintinie*, *Deido District Hospital*, and *Bonassama Hospital* could instantly alert donors through in-app notifications, emails, and SMS. This was especially useful in emergency scenarios such as traffic accidents, maternal hemorrhages, and surgeries.
- The implementation of **role-based access control (RBAC)** ensured system security and operational clarity. Donors, recipients, hospitals, and administrators each had separate interfaces and access rights. This feature minimized the risk of data breaches, a common concern in health informatics systems in Cameroon where digital privacy policies are still emerging.
- The platform allowed **donors to upload their health screening documents**—including HIV test results, hepatitis B screenings, and blood pressure logs—allowing hospitals to verify eligibility quickly. In the Douala context, where some health institutions still rely

on paper files and fragmented processes, this represented a major step toward modernized transfusion safety.

5.2 Recommendations

Given the above findings and considering the realities of the **Cameroonian healthcare ecosystem**, particularly in Douala, the following recommendations are made:

1. **Deploy Offline or SMS-Based Functionality:** To ensure the platform is accessible to hospitals in areas with limited internet connectivity, SMS-based blood requests and confirmations should be developed. This would allow hospitals in remote zones around Douala to continue functioning even during outages.
2. **Automate Donor Status Monitoring:** The platform should integrate automated reminders via SMS or WhatsApp, informing donors when they are eligible to donate again. Future versions can explore wearable device integration to monitor vital signs and donation readiness.
3. **Establish Government Collaboration:** The success and impact of LifeDrop can be amplified if endorsed by the **Cameroon Ministry of Public Health**. Integration with national hospital databases and blood banks would ensure real-time access to national blood availability and enable public-private cooperation.
4. **Launch Public Awareness Campaigns:** To increase donor registration and educate the public, LifeDrop should partner with local health NGOs, churches, universities (e.g., University of Douala, IUGET), and media houses for blood donation sensitization campaigns.
5. **Introduce Multi-language Support:** Cameroon is bilingual, with several regional dialects spoken. Providing the platform in **French, English, and selected local languages (e.g., Duala)** would improve inclusivity and usability for the diverse population of Douala.
6. **Strengthen Security Protocols:** Given the sensitivity of health information, LifeDrop should implement end-to-end encryption and comply with global health data standards such as **HIPAA or GDPR-equivalent** frameworks, adapted to the Cameroonian context.

5.3 Conclusion

The successful development and deployment of the **LifeDrop** platform underscores the critical role that **intelligent digital systems** can play in improving public health outcomes in **Cameroon**, especially in densely populated urban areas such as **Douala**, where health infrastructure is often overwhelmed and under-resourced.

LifeDrop directly addresses long-standing inefficiencies in the blood donation and transfusion ecosystem—ranging from delayed communication to donor-recipient mismatches and a lack of real-time coordination between stakeholders. By automating donor-recipient matching, simplifying hospital request workflows, and integrating with **mobile money platforms** common in Cameroon (such as MTN Mobile Money and Orange Money), LifeDrop reduces the time between request and delivery, potentially saving hundreds of lives in emergency situations.

Importantly, the system does not just serve hospitals or blood banks—it **empowers ordinary citizens** to participate actively in life-saving efforts. Donors are alerted instantly when their blood type is needed, and recipients or their families can easily make targeted requests. This **community-driven design** fosters a culture of civic engagement and public health responsibility that is essential for building a sustainable donation network.

In broader terms, LifeDrop proves that **digital innovation can serve as a catalyst for healthcare reform** in resource-constrained environments. The platform’s success offers a viable **model for national expansion**, especially in regions with weak emergency response mechanisms, fragmented donor databases, and limited healthcare digitization. It provides the framework upon which **policy-level transformations** can be built—such as national donor registries, real-time blood inventory systems, and integration with civil protection agencies.

Ultimately, LifeDrop is more than a software application—it is a blueprint for **modernizing healthcare delivery** in Cameroon. It bridges the gap between urgency and availability, technology and humanity, policy and practice. If scaled and supported strategically, it has the potential to redefine how Cameroon, and similar countries in Sub-Saharan Africa, manage critical healthcare logistics in the 21st century!

5.4 Suggestions for Further Study

To ensure that the innovations presented in the LifeDrop platform continue to evolve and meet the dynamic needs of healthcare delivery in Cameroon, especially in urban centers like **Douala**, further research and technical enhancement are recommended in the following areas:

1. Nationwide Scaling and Interoperability

Future studies should explore how LifeDrop can be scaled and standardized for **nationwide implementation**, connecting public and private health institutions across all ten regions of Cameroon. Interoperability research should focus on:

- **Integrating LifeDrop with government platforms** such as the National Blood Transfusion Service (NBTS) and regional hospital databases.
- Exploring **API-level compatibility** with systems used by NGOs like **Doctors Without Borders, the Red Cross, or the Cameroon Baptist Convention Health Services**.
- Developing a **unified national blood bank digital infrastructure** to enable real-time visibility into blood availability across cities like Yaoundé, Buea, Bamenda, and Garoua.

This would allow for better **resource optimization, inter-regional emergency support, and centralized health reporting** for policymaking.

2. Impact Assessment on Mortality Rates

While LifeDrop's features show promising results, a more **scientific assessment of its real-world impact** is necessary. Future research should:

- Collect data on **emergency response times** before and after LifeDrop's implementation in hospitals across Douala, such as Hôpital Laquintinie, Hôpital Général de Douala, and Bonassama District Hospital.
- Measure the **reduction in preventable deaths** caused by delayed blood availability, particularly among maternal patients, accident victims, and children with anemia or sickle cell disease.

- Use statistical tools such as **regression models and survival analysis** to correlate LifeDrop usage with patient outcomes, creating an evidence-based case for expansion.

3. Blockchain-Based Blood Tracking

The current version of LifeDrop ensures traceability through digital logging. However, future enhancements can explore the use of **blockchain technology** for:

- Creating a **tamper-proof ledger** for blood donations and transfusions, ensuring that all movements of blood units—from donor to hospital to recipient—are securely documented.
- Preventing **fraudulent activities** like blood mislabeling, black market sales, or counterfeit health documents.
- Allowing **auditable, decentralized transparency**, especially when coordinating with international health bodies or during humanitarian crises.

Such research could position Cameroon as a pioneer in **ethical and secure digital blood management systems** in Sub-Saharan Africa.

4. Integration with AI Predictive Models

A promising direction for future development is the integration of **artificial intelligence (AI)** and **machine learning** algorithms into LifeDrop’s core engine. Suggested studies include:

- Building **predictive models** using hospital records, accident reports (e.g., from Douala’s police or transport offices), seasonal trends (e.g., malaria or dengue spikes), and public events (festivals, holidays).
- Training AI to **forecast blood demand per region**, enabling hospitals to proactively prepare and coordinate donation drives.
- Studying how **data from climate patterns**, road safety statistics, and infectious disease reports can be included in the model to improve its accuracy and impact.

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APPENDIX

Appendix A: Screenshots of User Interfaces Figure A1: Donor Registration Page Figure A2: Hospital Registration Page Figure A3: Recipient Registration Page Figure A4: Create Blood Donation Request (Hospital) Figure A5: View Available Donors (Hospital Dashboard) Figure A6: Admin Dashboard Interface.

Appendix B: Sample Code Snippets

B1. Donor Registration (Node.js & MongoDB): `const newDonor = new Donor({ name, email, bloodGroup, city, phoneNumber, lastDonationDate }); await newDonor.save();`

B2. Creating Blood Request (Hospital): `const request = new BloodRequest({ hospitalId, bloodType, urgency, unitsNeeded, location }); await request.save();`

Appendix C: Interview Questions (Qualitative Data Collection) 1. Have you ever donated blood before? Why or why not? 2. What challenges have you experienced when looking for a blood donor or recipient? 3. How comfortable would you be using a mobile or web app to manage blood donations?

Appendix D: Survey Summary Target Participants: 5 hospital staff, 3 donors, 8 recipients Key Insight: 85% believe a mobile/web-based platform would improve emergency response.

Appendix E: System Technologies Used Backend: Node.js, Express.js Database: MongoDB Frontend: HTML, CSS, JavaScript Realtime Communication: Socket.IO Email Notification: Nodemailer + Gmail App Password Hosting: Render (Backend), Netlify (Frontend)

Appendix F: List of Abbreviations UI - User Interface WHO - World Health Organization HBM - Health Belief Model RBAC - Role-Based Access Control API - Application Programming Interface